외계행성 대기
Exoplanetary Atmospheres

Lecture 6: Radiative Transfer IV — Practical Solutions and the Two-Stream Approximation
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Based on: Chapters 5–6 of Seager, *Exoplanet Atmospheres* (2010), and Chapters 2–3 of Heng, *Exoplanetary Atmospheres: Theoretical Concepts and Foundations* (2017)

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Outline – Lecture 6

Simple Solutions and the Practical Checklist

Clouds and Atmospheric Retrieval

The Two-Stream Approximation

The Diffusion Approximation

Clouds and Monte Carlo Radiative Transfer

Summary

The boundary conditions: a two-point problem

- To further investigate the hidden complication, return to the boundary conditions. To solve for the intensity from this *first-order* ordinary differential equation we require two full boundary conditions, that is, an upper and a lower boundary condition on the *full* range $-1 \leq \mu \leq 1$.
- Yet the information we have is the stellar radiation incident on the planet and traveling *downward*, $-1 \leq \mu < 0$, at the top, and an estimate of the interior energy that we may convert to an *outward-going* intensity, $0 < \mu \leq 1$, at the bottom.
- Because the boundary conditions are not fully specified — half of the rays are known at each boundary — iterative techniques or a different formulation of the transfer equation are needed to solve for I:
  - Λ iteration: guess S, compute I by the formal solution, update J and S, repeat (slow when $\omega_0 \to 1$; accelerated versions exist);
  - Feautrier's method: combine $I(\mu)$ and $I(-\mu)$ into symmetric and antisymmetric parts, giving a second-order equation with one condition at each boundary;
  - two-stream and discrete-ordinate methods (Heng Chapter 3), moment methods with a closure (Eddington approximation, Chapter 6), and Monte Carlo simulations.
- The same structure appears in every transport problem with scattering: the solution at a point depends on the solution everywhere else.

Beer's law

The simplest solution discards the source function ($S = 0$): pure extinction of an external beam.

$$\int_{I_0}^{I} \frac{\mathrm{d}I}{I} = \int_0^{\tau} \frac{\mathrm{d}\tau'}{\mu} \quad \Longrightarrow \quad \boxed{I = I_0\, \mathrm{e}^{-\tau/|\mu|}}$$

(Heng writes $I_0 \mathrm{e}^{\tau/\mu}$ with his $\mu < 0$ for penetrating rays; the intensity decreases either way.) This is **Beer's law** (or the Beer–Lambert–Bouguer law): the intensity is diminished exponentially, $\tau = 1$ corresponding to one e-folding, i.e. 63% of the beam removed ($1/\mathrm{e} \simeq 37\%$ transmitted) when unit optical depth is traversed.

- The slant factor $1/|\mu|$: a beam entering at 60° from the normal traverses twice the optical depth of a normal beam. This is why the Sun is redder at sunset and why the transit chord is so opaque.
- Beer's law describes very well how *starlight* is absorbed on entering an exoplanet atmosphere, and it underlies transmission spectroscopy: the in-transit flux is the stellar flux times $\mathrm{e}^{-\tau_{\mathrm{chord}}(\nu)}$ integrated over the annulus.
- It is inadequate for the planet's *thermal emission*, which has both incoming and outgoing components and a source function that cannot be neglected.
- Heng Chapter 4 generalizes Beer's law to include non-isotropic scattering of the stellar beam.

Direct solution with pure absorption: a pair of layers

Beyond Beer's law, the equation can be solved directly for I only in the limit of *pure absorption*. Include coherent isotropic scattering, $S = (1-\omega_0)B + \omega_0 J$, and integrate between two points 1 and 2 ($\tau_2 > \tau_1$):

$$I_2 \mathrm{e}^{-\tau_2/\mu} - I_1 \mathrm{e}^{-\tau_1/\mu} = -\frac{1}{\mu}\int_{\tau_1}^{\tau_2} \left[\omega_0 J + (1-\omega_0)B\right] \mathrm{e}^{-\tau/\mu}\,\mathrm{d}\tau.$$

One cannot proceed unless $\omega_0 = 0$, because $J(\tau)$ is not known a priori. ("Coherent" means the wavelength is unchanged; spectral lines are better described by complete redistribution, but for bins wide compared to line widths coherent scattering is a decent approximation.)

Isothermal absorbing layer. Take $\omega_0 = 0$ and B, I_1, I_2 constant; label upward and downward radiation by $\uparrow$, $\downarrow$; integrate over the hemisphere and write fluxes as $F = \pi I$. Then

$$\boxed{F_{\uparrow 1} = F_{\uparrow 2}\,\mathscr{T} + \pi B(1-\mathscr{T}), \qquad F_{\downarrow 2} = F_{\downarrow 1}\,\mathscr{T} + \pi B(1-\mathscr{T})}$$

with the **transmission function**

$$\mathscr{T} \equiv 2\int_0^1 \mu\,\mathrm{e}^{-\Delta\tau/\mu}\,\mathrm{d}\mu = 2E_3(\Delta\tau) = (1-\Delta\tau)\mathrm{e}^{-\Delta\tau} + \Delta\tau^2 E_1(\Delta\tau), \qquad \Delta\tau \equiv \tau_2 - \tau_1,$$

where $E_n(x) = \int_1^\infty t^{-n}\mathrm{e}^{-xt}\,\mathrm{d}t$ are the exponential integrals (Heng, problem 2.8.5).

The transmission function

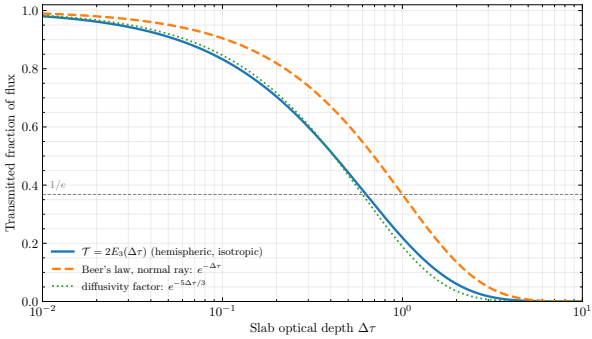


Figure 1: Hemispheric transmission $\mathscr{T} = 2E_3(\Delta\tau)$ of an isothermal slab compared with Beer's law for a normal ray and with the diffusivity-factor approximation $e^{-5\Delta\tau/3}$.

- $\mathscr{T}$ has a convenient range between 0 and 1. When $\mathscr{T} = 1$ no radiation is absorbed and no blackbody emission is produced: the incident flux passes right through the pair of layers, a completely transparent medium. When $\mathscr{T} = 0$ nothing passes through and the pair of layers radiates purely as a blackbody, $F = \pi B$.
- Because the flux is an average over all angles, and slant rays see a larger optical depth $\Delta\tau/\mu$, $\mathscr{T}$ falls faster than the normal-ray Beer's law $e^{-\Delta\tau}$: the slab reaches $1/e$ transmission at $\Delta\tau \simeq 0.6$, not at 1.

The transmission function: approximations and use

- **Diffusivity factor.** The classic approximation replaces the angle integral by a single ray at an effective angle, $\mathscr{T} \simeq \mathrm{e}^{-D\Delta\tau}$ with $D = 5/3$ ($\mu_{\mathrm{eff}} = 3/5$; Elsasser 1942). As the figure shows it is accurate to a few percent for all $\Delta\tau$, which is why single-angle (two-stream) methods work so well for thermal radiation.
- **Small $\Delta\tau$:** $\mathscr{T} \simeq 1 - 2\Delta\tau$ (not $1 - \Delta\tau$): a thin slab absorbs twice as much hemispheric flux as a normal ray would suggest, because the mean slant factor $\langle 1/\mu \rangle$ over the flux-weighted hemisphere is 2.
- **Large $\Delta\tau$:** $\mathscr{T} \simeq \mathrm{e}^{-\Delta\tau}(1 - \Delta\tau) + \Delta\tau^2 E_1 \to 2\mathrm{e}^{-\Delta\tau}/\Delta\tau$ asymptotically, dominated by the normal rays, the only ones that get through.
- **Pairs of layers.** Writing the solutions in this form describes the transfer of radiation from point 1 to point 2 and vice versa, using either location as the starting point or boundary condition. Formulating the problem as radiative transfer between *pairs of layers* is the basis of the two-stream treatment (Heng Chapter 3), and stacking many such layers is how a numerical longwave code computes the emergent flux for an arbitrary $T(\tau)$ profile.

Non-isothermal layers: a linear source function

The Planck function generally varies with optical depth. A direct solution still exists if B is expanded to first order about the layer boundaries (Heng, problem 2.8.5b):

$$B = \begin{cases} B_1 + B'(\tau - \tau_1), & \downarrow \text{ direction}, \\ B_2 + B'(\tau - \tau_2), & \uparrow \text{ direction}, \end{cases} \qquad B' \equiv \frac{\partial B}{\partial \tau} = \frac{B_2 - B_1}{\tau_2 - \tau_1},$$

which reproduces B_1 at τ_1 and B_2 at τ_2. The hemispherically integrated solutions become

$$F_{\uparrow 1} = F_{\uparrow 2}\mathscr{T} + \pi B_2(1 - \mathscr{T}) + \pi B' \left[\tfrac{2}{3}\left(1 - \mathrm{e}^{-\Delta\tau}\right) - \Delta\tau\left(1 - \tfrac{\mathscr{T}}{3}\right)\right],$$

$$F_{\downarrow 2} = F_{\downarrow 1}\mathscr{T} + \pi B_1(1 - \mathscr{T}) - \pi B' \left[\tfrac{2}{3}\left(1 - \mathrm{e}^{-\Delta\tau}\right) - \Delta\tau\left(1 - \tfrac{\mathscr{T}}{3}\right)\right].$$

- The extra term is the correction from the temperature gradient within the layer; it vanishes for $B' = 0$ and for $\Delta\tau \to 0$.

Eddington–Barbier and the formation of bands

- For a source function linear in optical depth, $S = a + b\tau$, the emergent-intensity integral gives *exactly*

$$I(0,\mu) = \frac{1}{\mu}\int_0^\infty (a+b\tau)\,\mathrm{e}^{-\tau/\mu}\,\mathrm{d}\tau = a + b\mu = S(\tau=\mu),$$

  the **Eddington–Barbier relation**: the emergent intensity equals the source function at optical depth μ along the ray. For a general S it remains a good approximation.

- **Limb darkening.** Looking at the disk center ($\mu = 1$) we see $S(\tau = 1)$; near the limb ($\mu \to 0$) we see $S(\tau \to 0)$. If T decreases outward, the limb is darker; a stellar limb darkening curve therefore measures $\mathrm{d}S/\mathrm{d}\tau$, and modeling it is essential for transit radii (Lecture 1).

- **Absorption bands.** In a strong molecular band κ_ν is large, $\tau_\nu = \mu$ is reached high up where T is low, and $I(0,\mu,\nu) = B(T_{\mathrm{high}})$ is small: the band appears in *absorption*. In a spectral window we see deeper, warmer layers and more flux.

- **Emission bands.** If T *increases* with height (a stratospheric inversion, as for ozone on Earth or TiO/VO on some hot Jupiters), the same bands appear in *emission*. The sign of the features in an emission spectrum thus diagnoses the sign of the temperature gradient.

- This is the whole basis of remote sensing of temperature structure: each frequency samples S, hence T, at its own $\tau_\nu \sim \mu$, i.e. its own pressure (Chapter 6 and Chapter 9).

Why the linear source function matters

- The "linear-in-τ" source function is the standard building block of numerical longwave radiative transfer, for example in the widely used two-stream scheme of Toon et al. (1989) and in most exoplanet forward models: within each layer B is interpolated linearly in τ, and the layer solutions are chained together from the bottom boundary to the top.
- Integrated over a semi-infinite atmosphere, it gives the **Eddington–Barbier relation for the flux**: for $S = a + b\tau$, $F(0) = 2\pi \int_0^1 \mu (a + b\mu)\, \mathrm{d}\mu = \pi(a + \frac{2}{3}b) = \pi S(\tau = \frac{2}{3})$. The emergent flux equals π times the source function at optical depth $2/3$ — the origin of the rule of thumb that the photosphere sits at $\tau \simeq 2/3$, and of the appearance of $2/3$ in grey-atmosphere temperature profiles.
- The same relation for the intensity gives $I(0, \mu) = S(\tau = \mu)$ (previous section), so the disk-center intensity samples $\tau = 1$ and the limb samples $\tau \to 0$: limb darkening measures $\mathrm{d}S/\mathrm{d}\tau$, i.e. the temperature gradient.
- With many thin layers, each isothermal or linear, the emergent flux of an arbitrary $T(\tau)$ profile can be built up layer by layer — exactly the forward model used in atmospheric retrievals (below), where speed matters because it is evaluated millions of times.

A practical checklist for radiative transfer calculations

What is needed to actually compute a spectrum (Heng, Section 2.5):

1. **Choose the form of the transfer equation** and the approximations (plane-parallel? scattering? two-stream or multi-stream?), check that the assumptions are plausible, and select the numerical method. Solutions are called *forward models*: they take abundances and opacities and produce a synthetic spectrum.
2. **Set up a model atmosphere** with a finite number of layers, each with its own T, P, and column mass, and compute the fluxes passing through each layer. The isothermal-layer solution above is a forward model in the pure-absorption limit; its inputs are the opacities, from which the optical depths, transmission functions, and fluxes of each layer follow.
3. **Decide the chemical composition** (Heng Chapter 7) and obtain the *opacities* of the atoms, molecules, and aerosols, typically from precomputed tables (Heng Chapter 5). Combining them, weighted by abundance, over T, P, and λ gives the opacity function of the atmosphere.
4. **Iterate to convergence**: opacities, optical depths, and temperatures are interdependent. Convergence is reached when the temperature of each layer stops changing (to within a numerical threshold).

The converged solution yields the flux at the top of the atmosphere (the synthetic spectrum) and the temperatures of all layers (the temperature–pressure profile).

Radiative equilibrium: local energy conservation

How are the temperatures and fluxes of the layers related? In the absence of atmospheric dynamics one hopes to solve for **radiative equilibrium**, which occurs when the net heat entering and leaving each layer vanishes. It is a statement of *local* energy conservation and originates from the first law of thermodynamics (Heng Chapter 9),

$$\rho c_P \frac{DT}{Dt} = \frac{DP}{Dt} + \rho Q,$$

with ρ the mass density, c_P the specific heat at constant pressure, and Q the heating term. Focusing on radiative heating (ignoring convection and conduction),

$$\rho Q = -\nabla \cdot \mathbf{F}_-,$$

where $\mathbf{F}_-$ is the wavelength-integrated *net* flux: the difference between the flux entering and exiting a layer, hence the minus subscript.

- Geometrically this makes sense: when radiation emanates from the layer, the divergence of the net flux is positive, which leads to cooling ($Q < 0$). The opposite happens when radiation enters the layer.
- The minus sign is crucial: if it is missing, one gets the unphysical situation of runaway cooling or heating.

Radiative equilibrium: the working equation

Ignoring work done on the system ($DP/Dt = 0$) and atmospheric dynamics ($D/Dt = \partial/\partial t$) gives the equation typically used to solve for radiative equilibrium in a one-dimensional model,

$$\boxed{\frac{\partial T}{\partial t} = -\frac{1}{\rho c_P}\frac{\partial F_-}{\partial z}} \qquad (z \text{ increasing with altitude}).$$

- The computational iteration between the fluxes and temperatures, which depend on each other, is performed until the temperature stops changing in time (in practice until $\partial T/\partial t$ falls below a threshold, e.g. 10^{-6}; an exact zero is impossible numerically). Then radiative equilibrium is formally obtained: $\mathrm{d}F_-/\mathrm{d}z = 0$, a constant net flux through every layer.
- In frequency-resolved form the same condition reads $\int_0^\infty \kappa_\nu (J_\nu - B_\nu)\,\mathrm{d}\nu = 0$ in each layer: the energy absorbed equals the energy emitted. This is what fixes $T(\tau)$ and therefore B in the source function.
- Strictly speaking, if a calculation does not enforce radiative equilibrium, then we should not consider it a bona fide radiative transfer calculation of an atmosphere.

Local versus global energy conservation

- **Radiative equilibrium** is a condition on *every layer*: the net radiative flux through the atmosphere is the same at all heights, so no layer is heating or cooling. It determines the temperature–pressure profile.
- **Global energy conservation** is a condition on the atmosphere *as a whole*: the energy entering (absorbed starlight plus interior heat) equals the energy exiting (thermal emission to space). This is the energy balance of Lecture 2, which determines T_{eff} but not the profile.
- The two are often confused. Global conservation is a *necessary but insufficient* condition for radiative equilibrium: an atmosphere can have the right total emission yet an internally inconsistent profile in which some layers heat and others cool. A retrieval that enforces only the global condition may return such a profile.
- In the presence of **dynamics** radiative equilibrium is replaced by radiative–dynamical equilibrium: the divergence of the radiative flux is balanced by advection and by convection (where the radiative gradient exceeds the adiabatic one, Heng Chapter 4 and Seager Chapter 9). Even then the global condition holds for the planet as a whole, averaged over time.
- Heng Chapter 4 casts the distinction in precise mathematical terms using the double-grey solutions.

Clouds: the outstanding puzzle

The formation of aerosols or condensates in a gas is an outstanding puzzle of astrophysics (brown dwarfs, protoplanetary disks, exoplanets) and of Earth's climate science. Heng uses cloud, haze, aerosol, and condensate interchangeably, aware of the subtle differences: hazes are usually photochemical products; clouds follow the thermodynamic condensation curves. Forming clouds from first principles requires:

- **Nucleation:** how the precursor seed particles form, on which further growth proceeds; a true theory predicts the shape, composition, and size distribution of the particles.
- **Self-consistent gas and solid chemistry:** as the gas is depleted to form solids, its atomic and molecular abundances adjust.
- **Self-consistent opacities and radiative transfer** for gas and cloud particles, with the absorption and scattering of each computed from first principles across λ, T, and P rather than by ad hoc parameters.
- **Dynamics:** to stay aloft, the terminal velocity of a particle under gravity must be balanced by upwelling flow; sedimentation removes condensates below the photosphere. Usually parametrized by an eddy mixing coefficient K_{zz}.

Forming the particles is hard; once formed, modeling their effect on the spectrum is not controversial. "Aerosol" is the neutral term: any particle described by its size, composition, and index of refraction (or extinction efficiency).

What clouds do to a spectrum

To lowest order the aerosol adds an extinction efficiency $Q_e(x)$ with $x = 2\pi r/\lambda$ (the fitting formula and Heng Fig. 2.2 above):

- For $x \gg 1$, Q_e is roughly constant: a *grey* continuum opacity. For $x \ll 1$ it is Rayleigh-like, $\propto x^4$.
- The parameter Q_0 is a proxy for composition: refractory species (silicates) have $Q_0 \approx 10$, volatile ones (water) $Q_0 \approx 40$–80. Particle radius has a somewhat larger influence than composition.
- A cloud particle influences absorption and scattering only at $\lambda \lesssim 2\pi r$. This generalizes to a size distribution.

Consequences for spectra. When the cloud optical depth approaches unity above the gas photosphere, spectral features and lines become *muted* and less sharp for $\lambda \lesssim 2\pi r$ — the flat transmission spectra of GJ 1214b and the weakened water bands of many hot Jupiters. But the same muting can be produced by *lower* abundances of the absorbing species, so clouds introduce a **degeneracy** in the interpretation of chemical abundances. To the next order, particles are described by ω_0 and g_0, layer and wavelength dependent, in a two-stream or multi-stream calculation.

Atmospheric retrieval: letting the data talk

Traditionally, when one builds a model of an atmosphere, one starts with a governing equation — of fluid dynamics, radiation, chemistry, or some combination — and a set of assumptions, and computes. Ideally the calculation predicts a measurable quantity that may be confronted by the observations, and a judgment is made on whether the model has represented Nature — but only against the preconceived notions we have about an atmosphere.

- **Atmospheric retrieval** tries to shed those notions by providing a framework in which one *inverts* a set of observables (a spectrum) to obtain the properties of the atmosphere: its thermal structure and chemical abundances.
- Origin: inversion techniques for remote-sensing observations of Earth (Rodgers 2000), later used in the planetary sciences for the Solar System bodies. There, a combination of satellite and in-situ data provides ground truth to anchor the retrieval. “We can simply let the data do the talking.”
- The output is not a single model but a *posterior distribution*: the range of atmospheres consistent with the data and their uncertainties.

Why retrieval is harder for exoplanets

- For exoplanets the faith in the data alone is less viable: we are dealing with remote sensing of *unresolved point sources*, and in-situ measurements are out of the question.
- The data are often sparse and fragmented, in the sense that one only sees incomplete facets of the atmosphere — the dayside but not the nightside, insufficient wavelength coverage, a handful of photometric bands — that do not allow unique interpretations.
- Letting the interpretation be guided entirely by the data, for example without enforcing radiative or chemical equilibrium, becomes foolhardy when the number of model parameters exceeds the number of data points. Our interpretation needs to be anchored by the laws of physics and chemistry to some degree.
- How much physics to impose is an ongoing debate: “free” retrievals with parametrized profiles and abundances versus “self-consistent” models with equilibrium chemistry and radiative equilibrium. The former are flexible but can wander into unphysical corners; the latter are constrained but may be wrong if the assumed physics is incomplete.
- JWST-quality spectra, with hundreds of spectral points, have shifted the balance: the data now have enough information to test the assumptions of the self-consistent models rather than merely to fit a few parameters.

Basic ingredients of a retrieval model

Augmenting the radiative-transfer checklist:

1. A way of describing the **temperature–pressure profile** with a set of parameters: either an ad hoc fitting function, or a simplified analytic solution of the radiative transfer equation (Heng Chapter 4, the double-grey solutions).
2. The **identities of the atoms and molecules** included (e.g. water, methane, carbon monoxide). Their relative abundances may be free parameters of the retrieval; alternatively one may enforce *chemical equilibrium*, in which case the molecular abundances are determined by the elemental abundances and only the latter are free (Heng Chapter 7).
3. For a fixed combination of parameter values, a **forward model** converts the opacities into a synthetic spectrum. This conversion is performed once for each parameter set; there is typically no attempt to solve for radiative equilibrium, as the iteration described earlier is not performed.

Scanning the parameter space

- Millions, if not billions, of combinations of these parameters are considered. Each synthetic spectrum is compared against the observed one and a goodness of fit (a likelihood, typically $\propto e^{-\chi^2/2}$) is computed for each model.
- In this manner a range of best-fit models is found, given the uncertainties on the data, and the *posterior distributions* of the molecular abundances and of the temperature profile may be computed. Correlations between parameters (the degeneracies of the next slides) show up as elongated or banana-shaped posteriors.
- In practice a minimization or sampling algorithm is required to scan this large, multidimensional parameter space efficiently and thoroughly:
  - **Optimal estimation** exploits the property that if the prior of an input quantity is Gaussian, the expressions used for minimization simplify (Rodgers 2000). A uniform prior is mimicked by a very wide Gaussian.
  - **Monte Carlo methods** (Markov-chain Monte Carlo, nested sampling) allow general priors and have been implemented in various flavors; they are now standard in exoplanet retrievals.
- Because the forward model is evaluated millions of times, its speed is critical: this is why retrievals use fast layer-by-layer pure-absorption solutions with precomputed opacities rather than full radiative-equilibrium iterations.

Retrieval pitfalls (1)

A retrieval model is only as good as the assumptions and techniques used. Some of the possible pitfalls:

- **Degeneracies.** The spectral lines of different molecules inevitably overlap across wavelength, so multiple combinations of relative abundances may lead to the same net absorption at a given wavelength. Aerosols introduce another degeneracy: they diminish the strength of spectral lines, an effect that may be compensated by increasing the abundances of the molecules responsible. Any good retrieval quantifies these degeneracies formally and systematically (correlated posteriors).
- **Being misled by your priors.** The posteriors depend on the priors assumed. Example: the carbon-to-oxygen ratio estimated as

$$\mathrm{C/O} \approx \frac{\tilde{n}_{\mathrm{CO}} + \tilde{n}_{\mathrm{CH_4}} + \tilde{n}_{\mathrm{CO_2}}}{\tilde{n}_{\mathrm{CO}} + \tilde{n}_{\mathrm{H_2O}} + 2\tilde{n}_{\mathrm{CO_2}}},$$

  with $\tilde{n}$ the mixing ratios (abundances normalized by H_2). If CO dominates, C/O ≈ 1; if CO_2 dominates, C/O ≈ 0.5; if CH_4 or H_2O dominate, C/O $\gg 1$ or $\ll 1$. So even uniform priors on the molecules produce two artificial peaks in the prior distribution of C/O.

Retrieval pitfalls (2)

- **Local versus global energy conservation.** In the ensemble of models it is natural to discard those whose emergent thermal emission exceeds the energy input from the star (for negligible internal heat). That is simply global energy conservation: you cannot produce more than what you put in. What is less obvious is *local* energy conservation, radiative equilibrium: the net heating or cooling of each layer must be the same across layers, in the absence of circulation, and most retrievals do not enforce it.
- **Not everything is chemically possible.** In chemical equilibrium not every combination of relative abundances is possible: each layer has its own T and P, and the mixing ratios take specific values given the elemental abundances. One is not free to specify them as independent parameters. When dynamical mixing is faster than chemistry the mixing ratios depart from local equilibrium — but again not arbitrarily: with the *quenching* approximation (Heng Chapter 6) they originate from a deeper layer that *is* in equilibrium. Example: CO_2 in hot ($\gtrsim 1000\,\mathrm{K}$) hydrogen-dominated atmospheres is never more abundant than CO or H_2O unless the metallicity is implausibly high.

Retrieval pitfalls (3)

- **Opacities: the devil is in the details.** Across the UV to infrared one must compute the shapes of millions to billions of lines, for one molecule at one (T, P). Line-by-line calculations sample all of them without approximation, a formidable task at $\sim 1000\,\mathrm{K}$; binning techniques such as the k-distribution method speed this up at the cost of approximations (Heng Chapter 5). Hidden in the details is the pressure broadening of the far line wings, an unsolved physics problem. And photometric observations integrate over broad bandpasses, so many combinations of molecules can produce the same opacity across the band. When published papers omit these details, exercise skepticism — one of the most precious instruments in a scientist's toolkit.
- **Numerical convergence.** Running a retrieval requires discretizing space (the T–P profile), wavelength (the opacity function), and the set of species; to finish in reasonable time one often uses coarse resolution along one axis. Determining whether the resolution is sufficient requires resolution tests — thankless but necessary. As a rule of thumb (attributed by Heng to Scott Tremaine), always adopt the second-highest resolution if the highest one attains convergence.

What is the two-stream approximation?

Radiation in an atmosphere travels in all directions, is absorbed, re-emitted and scattered many times, possibly changing wavelength. Heng's strategy is to strip the problem down in two steps.

- **Physical basis.** In a *shallow* atmosphere (vertical extent $\ll R_p$) horizontal transport can be neglected, and radiation effectively flows only **up** and **down**. This holds for Earth, Mars, Titan and Venus, and reasonably for hot Jupiters, whose model domains are set by how deep starlight penetrates.
- **Mathematical basis.** The transfer equation has two independent variables, τ and μ. Integrating separately over the upward and the downward hemisphere removes μ and leaves two ordinary differential equations in τ alone.
- **The price.** Integration always produces one more unknown than equations. The system is closed by assuming ratios between angular moments — the **Eddington coefficients**. Different choices are different **closures**, and a careless choice can violate energy conservation.

Separate two-stream calculations are often done for starlight (shortwave) and for thermal emission (longwave), each with its own coefficients; this is the basis of the temperature–pressure profiles of the next lecture.

The starting equation and the moments

Heng's transfer equation with non-isotropic but **coherent** scattering (no change of wavelength), Eq. (3.1):

$$\mu \frac{\partial I}{\partial \tau} = I - \omega_0 \int_{4\pi} P I \, \mathrm{d}\Omega' - (1-\omega_0) B, \qquad \int_{4\pi} P \, \mathrm{d}\Omega = 1.$$

(Heng normalizes P to unity, so his P is our $P/4\pi$.) Define hemispheric moments of the intensity, with $\mathrm{d}\Omega = \mathrm{d}\mu \, \mathrm{d}\phi$:

$$J_\uparrow = \int_0^{2\pi} \int_0^1 I \, \mathrm{d}\mu \, \mathrm{d}\phi, \quad J_\downarrow = \int_0^{2\pi} \int_{-1}^0 I \, \mathrm{d}\mu \, \mathrm{d}\phi, \quad F_\uparrow = \int_0^{2\pi} \int_0^1 \mu I \, \mathrm{d}\mu \, \mathrm{d}\phi, \quad F_\downarrow = \int_0^{2\pi} \int_{-1}^0 \mu I \, \mathrm{d}\mu \, \mathrm{d}\phi.$$

- $J = J_\uparrow + J_\downarrow$ is Heng's "total intensity", i.e. 4π times the *mean* intensity used by Seager.
- $F_+ = F_\uparrow + F_\downarrow$ is the **total flux** and $F_- = F_\uparrow - F_\downarrow$ the **net flux**.[1]
- The analogy is with fluids: density and momentum are the moments of the Boltzmann equation.

[1] Although the integral for $F_\downarrow$ runs over $\mu < 0$, in the two-stream equations $F_\downarrow$ is used as a positive magnitude (an opaque isothermal layer gives $F_\downarrow = \pi B$); only then is $F_- = F_\uparrow - F_\downarrow$ the net upward energy flux.

Isotropic scattering: one equation, three unknowns

For isotropic scattering, $P = 1/4\pi$, the integral becomes $\omega_0 J/4\pi$:

$$\mu \frac{\partial I}{\partial \tau} = I - \frac{\omega_0 J}{4\pi} - (1-\omega_0) B.$$

Introduce a slant optical depth $\tau' = \tau/\bar{\mu}_\uparrow$, with $\bar{\mu}_\uparrow$ a characteristic cosine of the upward hemisphere, and integrate over that hemisphere:

$$\frac{1}{\bar{\mu}_\uparrow} \frac{\partial F_\uparrow}{\partial \tau'} = J_\uparrow - \frac{\omega_0 J}{2} - 2\pi(1-\omega_0) B.$$

- One equation, three unknowns ($F_\uparrow$, $J_\uparrow$, $J_\downarrow$). The downward hemisphere adds an equation but also $F_\downarrow$: the system stays *under-determined*.
- This is generic for moment methods: the equation for the n-th moment always contains the $(n+1)$-th.
- The factor $\bar{\mu}$ is book-keeping: it keeps track of the minus signs in the downward hemisphere and drops out of the final solutions.

Closing the system: the Eddington coefficients

Assume the *ratios* of fluxes to total intensities are constants:

$$\varepsilon_\uparrow \equiv \frac{F_\uparrow}{J_\uparrow}, \qquad \varepsilon_\downarrow \equiv \frac{F_\downarrow}{J_\downarrow}, \qquad \varepsilon \equiv \frac{F_+}{J}.$$

There is no physical reason for these ratios to be constant; it is a closure that makes progress possible. Assuming the two hemispheres are symmetric, $\bar{\mu} = \bar{\mu}_\uparrow = -\bar{\mu}_\downarrow$ and $\varepsilon' = \varepsilon_\uparrow = \varepsilon_\downarrow$, gives

$$\boxed{\frac{\partial F_\uparrow}{\partial \tau'} = \gamma_\mathrm{a} F_\uparrow - \gamma_\mathrm{s} F_\downarrow - \gamma_\mathrm{B} B, \qquad \frac{\partial F_\downarrow}{\partial \tau'} = -\gamma_\mathrm{a} F_\downarrow + \gamma_\mathrm{s} F_\uparrow + \gamma_\mathrm{B} B}$$

$$\gamma_\mathrm{a} = \bar{\mu}\left(\frac{1}{\varepsilon'} - \frac{\omega_0}{2\varepsilon}\right), \qquad \gamma_\mathrm{s} = \frac{\bar{\mu}\,\omega_0}{2\varepsilon}, \qquad \gamma_\mathrm{B} = 2\pi\bar{\mu}\,(1-\omega_0).$$

- Two coupled, first-order, linear ODEs with constant coefficients.
- γ_s is the **coupling**: scattering turns upward flux into downward flux and vice versa. Without scattering ($\gamma_\mathrm{s} = 0$) the two streams are independent.

Fixing the coefficients with two physical limits

(1) Pure scattering must not heat. Radiative equilibrium (Lecture 3) requires $\partial F_-/\partial z = 0$. With $\omega_0 = 1$ ($\gamma_\mathrm{B} = 0$), subtracting the two equations gives $\partial F_-/\partial \tau' = (\gamma_\mathrm{a} - \gamma_\mathrm{s})F_+$. Since scattering is coherent, this must vanish at *each* wavelength, while $F_+ \neq 0$:

$$\gamma_\mathrm{a} = \gamma_\mathrm{s} \quad \Longrightarrow \quad \varepsilon = \varepsilon'.$$

A single Eddington coefficient remains, the **first Eddington coefficient** ε.

(2) An opaque isothermal absorber must emit πB per hemisphere. For $\omega_0 = 0$ and $\Delta\tau \to \infty$ the solutions give $F_\uparrow = F_\downarrow = \gamma_\mathrm{B} B/\gamma_\mathrm{a}$. Evaluating $\int \mu B \,\mathrm{d}\Omega$ over a hemisphere gives πB, so $\gamma_\mathrm{B}/\gamma_\mathrm{a} = \pi$ and

$$\boxed{\varepsilon = \frac{1}{2}} \qquad \gamma_\mathrm{a} = \bar{\mu}\,(2 - \omega_0), \quad \gamma_\mathrm{s} = \bar{\mu}\,\omega_0, \quad \gamma_\mathrm{B} = 2\pi\bar{\mu}\,(1 - \omega_0).$$

There is *no freedom* left: the coefficients and ε are not independent knobs, a frequent misconception.

The diffusivity factor

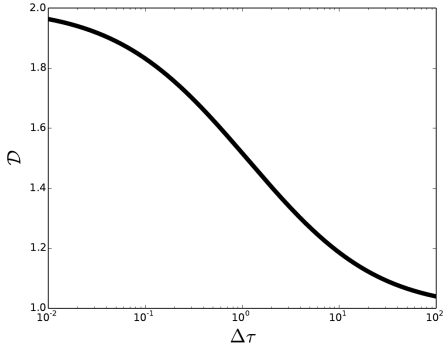


Figure 2: The diffusivity factor that makes the two-stream transmission exact for a purely absorbing isothermal layer. Heng (2017), Fig. 3.1.

In the purely absorbing isothermal case the two-stream transmission function is $\mathscr{T} = \mathrm{e}^{-\Delta\tau/\varepsilon} = \mathrm{e}^{-\mathscr{D}\Delta\tau}$, with the **diffusivity factor** $\mathscr{D} = 1/\varepsilon = 2$.

The exact transmission of diffuse light through a layer is $\mathscr{T} = 2E_3(\Delta\tau)$ (Lecture 3). Equating the two,

$$\mathscr{D} = -\frac{\ln\left[(1-\Delta\tau)\mathrm{e}^{-\Delta\tau} + \Delta\tau^2 E_1(\Delta\tau)\right]}{\Delta\tau} = -\frac{\ln 2E_3(\Delta\tau)}{\Delta\tau}.$$

- $\mathscr{D} \to 2$ for $\Delta\tau \ll 1$: the two-stream result is exact if the atmosphere is cut into optically thin layers.
- $\mathscr{D} \to 1$ for thick layers, where only nearly vertical rays get through.
- The popular $\mathscr{D} = 1.66$ was tuned for Earth. Heng's advice: do not treat $\mathscr{D}$ as a free parameter; use the value the closure implies.

General solution with isotropic scattering

Adding and subtracting the pair gives equations for the total and net fluxes:

$$\frac{\partial F_+}{\partial \tau'} = (\gamma_a + \gamma_s) F_-, \qquad \frac{\partial F_-}{\partial \tau'} = (\gamma_a - \gamma_s) F_+ - 2\gamma_B B,$$

which combine into one second-order equation,

$$\frac{\partial^2 F_+}{\partial \tau'^2} - (\bar{\mu}\mathscr{D})^2 F_+ = -2\gamma_B(\gamma_a + \gamma_s) B, \qquad \mathscr{D} = \frac{[(\gamma_a + \gamma_s)(\gamma_a - \gamma_s)]^{1/2}}{\bar{\mu}}.$$

- Within a layer the Planck function varies. Keep the first two terms of a Taylor series, $B = B_j + B'(\tau - \tau_j)$ with $B' \approx (B_2 - B_1)/(\tau_2 - \tau_1)$ — the linear source function again (Lecture 3).
- The solution is exponentials $\mathrm{e}^{\pm\alpha\tau'}$ plus a particular part $\pi(B \pm B'/2)$:

$$F_\uparrow = A_1 \zeta_+ \mathrm{e}^{\alpha\tau'} + A_2 \zeta_- \mathrm{e}^{-\alpha\tau'} + \pi\left(B + \frac{B'}{2}\right), \quad F_\downarrow = A_1 \zeta_- \mathrm{e}^{\alpha\tau'} + A_2 \zeta_+ \mathrm{e}^{-\alpha\tau'} + \pi\left(B - \frac{B'}{2}\right),$$

  with the **coupling coefficients** $\zeta_\pm = \frac{1}{2}\left[1 \pm (1-\omega_0)^{1/2}\right]$.
- A_1, A_2 follow from the boundary conditions: $F_{\uparrow 2}$ entering from below and $F_{\downarrow 1}$ from above (layer 1 sits above layer 2).

The layer-to-layer solutions

Eliminating A_1, A_2 (Heng Eq. 3.37 and problem 3.8.2), the outgoing flux at the top of a layer is

$$F_{\uparrow 1} = \frac{1}{(\zeta_- \mathscr{T})^2 - \zeta_+^2} \Big[(\zeta_-^2 - \zeta_+^2)\, \mathscr{T} F_{\uparrow 2} - \zeta_- \zeta_+ (1 - \mathscr{T}^2) F_{\downarrow 1} + (\text{Planck terms}) \Big], \qquad \mathscr{T} = \mathrm{e}^{-\mathscr{D}\Delta\tau},$$

and $F_{\downarrow 2}$ follows by symmetry.

- Every non-flux quantity ($\mathscr{T}$, $\zeta_\pm$) is dimensionless and lies between 0 and 1: numerically well behaved.
- $\zeta_\pm$ act as *weights*: how much of the light entering from below is transmitted ($\propto \mathscr{T}$) and how much of the light entering from above is reflected back ($\propto \zeta_- \zeta_+$).
- A model atmosphere is a stack of layers with boundary conditions at the top (starlight) and bottom (interior heat). The solutions are applied to each pair of layers and iterated until radiative equilibrium is reached.

Pure absorption ($\zeta_+ = 1$, $\zeta_- = 0$): $F_{\uparrow 1} = \mathscr{T} F_{\uparrow 2} + \pi(B_{1+} - B_{2+}\mathscr{T})$. An opaque layer ($\mathscr{T} = 0$) shows only its own top black body; a transparent one ($\mathscr{T} = 1$) passes the boundary flux plus a term that exists only if the layer is not isothermal.

Test case: a scattering slab

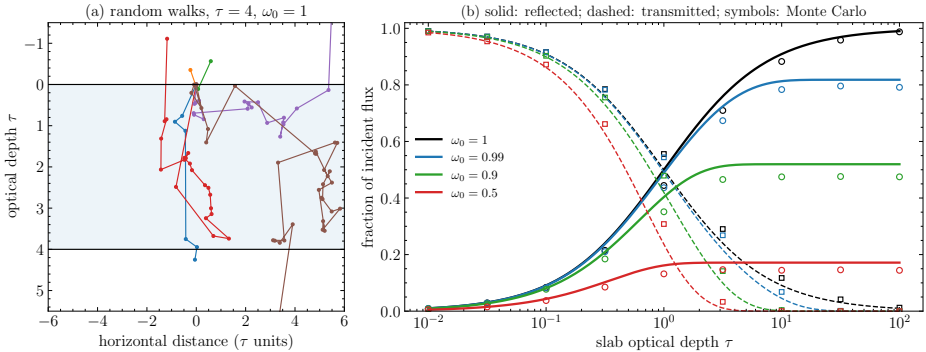


Figure 3: A homogeneous slab with isotropic scattering, lit from above by diffuse light, no emission. Lines: two-stream (hemispheric closure); symbols: Monte Carlo. Script `scripts/make_twostream_slab.py`.

Setting $B = 0$, $F_{\uparrow 2} = 0$: $R = \dfrac{\zeta_- \zeta_+ (1 - \mathscr{T}^2)}{\zeta_+^2 - \zeta_-^2 \mathscr{T}^2}$, $T_{\text{slab}} = \dfrac{(\zeta_+^2 - \zeta_-^2)\mathscr{T}}{\zeta_+^2 - \zeta_-^2 \mathscr{T}^2}$; a semi-infinite atmosphere reflects $R_\infty = \zeta_- / \zeta_+$.

What the test case teaches

- **Absorption caps the albedo.** However thick the slab, $R_\infty = \zeta_-/\zeta_+ = \dfrac{1-\sqrt{1-\omega_0}}{1+\sqrt{1-\omega_0}}$: 0.82 for $\omega_0 = 0.99$, 0.52 for 0.9, 0.17 for 0.5. Even 1% absorption per scattering removes a fifth of the light, because a reflected photon has scattered many times.
- **Conservative scattering** ($\omega_0 = 1$) needs its own solution (problem 3.8.4): $R = \tau/(1+\tau)$, $T_{\text{slab}} = 1/(1+\tau)$. Transmission falls only as $1/\tau$, not exponentially: photons diffuse through. The general formulas give $0/0$ here — Heng's caveat that "purely scattering" is not "purely reflecting".
- **Accuracy.** Against the Monte Carlo result the two-stream values are off by $\sim$ 10–20% at $\tau \sim 1$ (e.g. $\omega_0 = 1$: 0.50 versus 0.45). This is the typical price of two streams, and why multi-stream methods are used when scattering details matter.
- **Random walk.** In panel (a) photons wander far sideways before escaping; the number of steps grows as $\sim \tau^2$ (Lecture 3). That is the diffusion limit, discussed below.

The scattering phase function

Non-isotropic scattering can be handled *without* specifying P, using only two general properties.

1. P depends only on the **scattering angle** $\theta'' = \theta' - \theta$, with

$$\mu'' = \cos\theta'' = \mu'\mu + (1-\mu'^2)^{1/2}(1-\mu^2)^{1/2}\cos(\phi' - \phi).$$

   μ'' is unchanged if both $\mu \to -\mu$ and $\mu' \to -\mu'$, so P is invariant under this double sign flip.

2. P integrates to one in any of the three angular frames: $\int P\,\mathrm{d}\Omega = \int P\,\mathrm{d}\Omega' = \int P\,\mathrm{d}\Omega'' = 1$.

The usual fitting function is the **Henyey–Greenstein** phase function,

$$P = \frac{1}{4\pi}\,\frac{1-g_0^2}{\left(1+g_0^2-2g_0\cos\theta''\right)^{3/2}},$$

where g_0 is an input parameter; $g_0 = 0$ recovers $P = 1/4\pi$. In the derivation that follows, g_0 *emerges* as an integral over P.

Moments with non-isotropic scattering

Multiply Eq. (3.1) by a function $H(\theta)$ and integrate over all angles.

- $H = 1$ gives the net-flux equation,

$$\frac{\partial F_-}{\partial \tau'} = \bar{\mu}\,(1-\omega_0)(J - 4\pi B),$$

  already of the isotropic form: scattering changes directions, not the energy budget.

- $H = \mu$: writing $\mu = \cos(\theta' - \theta'')$, the cross term is odd under the double sign flip and vanishes; what remains is the first moment of the phase function,

$$g_0 \equiv \int_{4\pi} \mu''\, P\, \mathrm{d}\Omega'',$$

  the **asymmetry factor** ($g_0 > 0$ forward, < 0 backward, 0 isotropic). It is to P what the flux is to I. With the second moments $K_{\uparrow,\downarrow} = \int \mu^2 I\, \mathrm{d}\Omega$ over each hemisphere and $K_- = K_\uparrow - K_\downarrow$,

$$\frac{\partial K_-}{\partial \tau'} = \bar{\mu}\, F_+\,(1 - \omega_0 g_0).$$

  K/c is the radiation pressure.

The second Eddington coefficient and the hemispheric closure

Two equations, four unknowns (J, F_-, F_+, K_-). The first Eddington coefficient links J and F_+ ($\varepsilon = 1/2$); one more is needed. Define $\varepsilon_2 \equiv K_-/F_+$ and demand that the isotropic equations be recovered, which fixes $\varepsilon_2 = F_+/2F_-$ and gives

$$\frac{\partial F_+}{\partial \tau'} = 2\bar{\mu}\, F_- \,(1 - \omega_0 g_0).$$

Comparing with the isotropic pair yields the general coefficients

$$\boxed{\gamma_{\mathrm{a}} = \bar{\mu}\,[2 - \omega_0(1+g_0)], \quad \gamma_{\mathrm{s}} = \bar{\mu}\,\omega_0(1-g_0), \quad \gamma_{\mathrm{B}} = 2\pi\bar{\mu}\,(1-\omega_0)}$$

and coupling coefficients $\zeta_\pm = \frac{1}{2}\left[1 \pm \sqrt{(1-\omega_0)/(1-\omega_0 g_0)}\right]$.

- This is the **hemispheric** (hemi-isotropic) closure, here *derived* rather than assumed.
- The solutions keep the same form as for isotropic scattering, with $\mathscr{T} = \mathrm{e}^{-\mathscr{D}\Delta\tau}$ and $\mathscr{D} = 2\sqrt{(1-\omega_0)(1-\omega_0 g_0)}$.
- Forward scattering acts like weaker scattering: γ_{s} depends on $\omega_0(1-g_0)$, so a strongly forward-scattering cloud is more transparent than its ω_0 alone suggests.

Different closures

closure	$\gamma_a/\bar{\mu}$	$\gamma_s/\bar{\mu}$	$\gamma_B/\bar{\mu}$
hemispheric	$2-\omega_0(1+g_0)$	$\omega_0(1-g_0)$	$2\pi(1-\omega_0)$
Eddington	$\frac{1}{4}\left[7-\omega_0(4+3g_0)\right]$	$-\frac{1}{4}\left[1-\omega_0(4-3g_0)\right]$	$2\pi(1-\omega_0)$
quadrature	$\frac{\sqrt{3}}{2}\left[2-\omega_0(1+g_0)\right]$	$\frac{\sqrt{3}}{2}\,\omega_0(1-g_0)$	$\sqrt{3}\,\pi(1-\omega_0)$

Two tests every closure must pass:

1. **Energy conservation for pure scattering:** $\partial F_-/\partial\tau \propto \gamma_a-\gamma_s \propto 1-\omega_0$. All three pass.
2. **Correct black body for an opaque, isothermal absorber:** the figure of merit $f_\infty \equiv \gamma_B/[\pi(\gamma_a-\gamma_s)]$ must equal 1 (each hemisphere emits πB). Hemispheric and quadrature pass.

Why the Eddington closure fails the second test

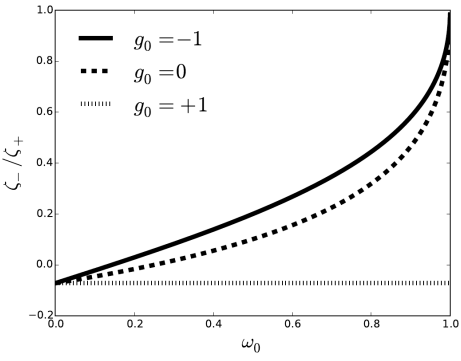


Figure 4: Spurious enhancement or reduction factor ζ_-/ζ_+ of the Eddington closure for an opaque layer. Heng (2017), Fig. 3.2.

- At $\omega_0 = 0$ (no scattering at all) the Eddington closure gives $\gamma_s = -\bar{\mu}/4 \neq 0$: upward rays are turned into downward ones, i.e. *reflection without scatterers*.
- Then $\zeta_\pm = \frac{1}{2}[1 \pm \sqrt{(\gamma_a - \gamma_s)/(\gamma_a + \gamma_s)}]$ gives $\zeta_-/\zeta_+ \approx -0.07$ and

$$f_\infty = \frac{\gamma_B}{\pi(\gamma_a - \gamma_s)}\left(1 - \frac{\zeta_-}{\zeta_+}\right) \approx 1.07,$$

  a spurious $\sim 10\%$ excess of black-body flux.
- In general both reflected and emitted fluxes are distorted by ζ_-/ζ_+, by amounts that depend on ω_0 and g_0 and compound from layer to layer.

Heng's point: a model meant to scan broad parameter space must be physically consistent everywhere, not merely tuned to work for one planet. (Goody & Yung's textbook uses the Eddington closure.)

Deep in the atmosphere: radiative diffusion

When the photon mean free path is much shorter than the scale height, conditions change little over one free path. Expand the source function about the local depth τ (Seager Eqs. 6.48–6.51):

$$S(\tau') \approx B(\tau) + \frac{\mathrm{d}B}{\mathrm{d}\tau}(\tau' - \tau) \quad \Longrightarrow \quad I(\tau, \mu) \approx B(\tau) + \mu \frac{\mathrm{d}B}{\mathrm{d}\tau}.$$

- The field is almost isotropic and Planckian, with a small anisotropy $\propto \mu$ that carries the flux; as $\tau \to \infty$, $I = S = B$.
- Its moments give $K = J/3$ exactly — Heng's **third Eddington coefficient** $\varepsilon_3 = K/J = 1/3$ — and a flux

$$F = 2\pi \int_{-1}^{1} \mu I \, \mathrm{d}\mu = \frac{4\pi}{3} \frac{\mathrm{d}B}{\mathrm{d}\tau} \qquad \text{(upward positive, } \tau \text{ increasing downward)}.$$

- Here the two streams merge: net and total fluxes are no longer distinct, and the two-stream picture gives way to diffusion. Scattering is effectively isotropic ($g_0 = 0$) and $J = 4\pi B$.

The equation of radiative diffusion

With $\mathrm{d}\tau = -\kappa\,\mathrm{d}z$ (Seager: κ per unit length) and integrating over frequency,

$$F_\nu = -\frac{4\pi}{3\kappa_\nu}\frac{\partial B_\nu}{\partial T}\frac{\mathrm{d}T}{\mathrm{d}z} \quad\Longrightarrow\quad \boxed{F = -\frac{16\,\sigma_{\mathrm{SB}}T^3}{3\,\kappa_{\mathrm{R}}}\frac{\mathrm{d}T}{\mathrm{d}z}}$$

where the frequency average that makes this exact is the **Rosseland mean** (Lecture 3),

$$\frac{1}{\kappa_{\mathrm{R}}} = \frac{\int_0^\infty \kappa_\nu^{-1}\,(\partial B_\nu/\partial T)\,\mathrm{d}\nu}{\int_0^\infty (\partial B_\nu/\partial T)\,\mathrm{d}\nu}, \qquad \int_0^\infty \frac{\partial B_\nu}{\partial T}\,\mathrm{d}\nu = \frac{4\sigma_{\mathrm{SB}}T^3}{\pi}.$$

- The form of Fourier's law for heat conduction: flux = conductivity × temperature gradient, with the **radiative conductivity** $16\sigma_{\mathrm{SB}}T^3/3\kappa_{\mathrm{R}}$. Heat flows down the temperature gradient.
- Heng uses opacity per unit mass: replace κ_{R} by $\rho\kappa_{\mathrm{R}}$. With hydrostatic equilibrium, $\mathrm{d}P = -\rho g\,\mathrm{d}z$,

$$F = \frac{16\,\sigma_{\mathrm{SB}}\,g\,T^3}{3\,\kappa_{\mathrm{R}}}\frac{\mathrm{d}T}{\mathrm{d}P}.$$

Why a harmonic mean

Three ways to average 1 and 100 (the Pythagorean means):

$$\text{arithmetic } \frac{1+100}{2} = 50.5, \qquad \text{geometric } (1\times 100)^{1/2} = 10, \qquad \text{harmonic } \frac{2}{1+1/100} \approx 2.$$

- The arithmetic mean weights both equally; the geometric mean averages in the logarithm; the harmonic mean is dominated by the *smallest* value.
- In an optically thick medium radiation leaks out through the **windows** — the wavelengths of lowest opacity — just as current in parallel resistors flows through the smallest resistance. The Rosseland mean is a harmonic mean for this reason.
- Hence a small amount of an opacity source that fills a window (a continuum absorber, a cloud) can change κ_{R} and the deep thermal structure far more than a strong line in an already opaque band.

Milne's solution

Deep in the atmosphere the flux is carried outward by radiation and, in radiative equilibrium, is constant with depth: set it to the internal heat flux, $F = \sigma_{\rm SB} T_{\rm int}^4$. Integrating $\sigma_{\rm SB} T_{\rm int}^4 = (16\sigma_{\rm SB} T^3/3)\,{\rm d}T/{\rm d}\tau_R$ over the Rosseland optical depth τ_R,

$$\boxed{T^4 = \frac{3}{4} T_{\rm int}^4 \,(\tau_R + C)}$$

with $C \sim 1$ a constant of integration ($C = 2/3$ in the Eddington approximation).

- This is **Milne's solution**, first derived for stars. At $\tau_R \approx 1$ the observer measures $T \approx T_{\rm int}$: for the Sun, $T_{\rm int} \approx 5800\,{\rm K}$ is the photospheric temperature.
- Temperature rises as $\tau_R^{1/4}$ with depth. In an irradiated planet this internal part is joined by the absorbed starlight; combining the two is the job of the next lecture.

Is the diffusion approximation valid? (Heng, problem 3.8.5)

Write the diffusion law as $F = -l_{\rm diff}\,\partial(\sigma_{\rm SB}T^4)/\partial z$, so $l_{\rm diff} = 4/(3\rho\kappa_{\rm R})$, and compare it with the scale height $H = k_{\rm B}T/mg$.

- **Deep hot-Jupiter atmosphere:** $P = 100\,\mathrm{bar}$, $T = 1000\,\mathrm{K}$, $m = 2m_{\rm H}$, $\kappa_{\rm R} = 0.01\,\mathrm{cm^2\,g^{-1}}$, $g = 10^3\,\mathrm{cm\,s^{-2}}$. Then $\rho = Pm/k_{\rm B}T = 2.4\times10^{-3}\,\mathrm{g\,cm^{-3}}$, $l_{\rm diff} \approx 0.55\,\mathrm{km}$, $H \approx 410\,\mathrm{km}$, and $H/l_{\rm diff} \approx 750$: **diffusion is excellent**.
- **Earth:** $P = 1\,\mathrm{bar}$ gives a column mass $P/g \approx 1.0\times10^3\,\mathrm{g\,cm^{-2}}$ (10 tonnes per $\mathrm{m^2}$). An infrared optical depth of 2 implies $\kappa \approx 2\times10^{-3}\,\mathrm{cm^2\,g^{-1}}$. With $\rho = 1.2\times10^{-3}\,\mathrm{g\,cm^{-3}}$, $l_{\rm diff} \approx 6\,\mathrm{km}$ while $H \approx 8\,\mathrm{km}$: $H/l_{\rm diff} \approx 1.5$, so **diffusion fails**.

Earth's lower atmosphere is only marginally opaque in the infrared; its thermal structure needs full radiative transfer (plus convection), not a diffusion law.

Radiative transfer in cloudy atmospheres

The equations above apply to clouds too; the practical difficulty is the sharp jump in opacity at the edges of an optically thick cloud. Approaches in the literature:

- Put the lower boundary *at* the cloud top, arguing that photons scatter in the cloud until they are absorbed.
- Average the emergent intensity of clear and cloudy lines of sight (partial cloud cover, clouds at several heights).
- Replace the cloud edge by an exponential fall-off in density, for numerical convenience.
- Represent the cloud by an analytic reflection law instead of solving the transfer inside it.

For realistic, patchy, three-dimensional clouds the most accurate tool is **Monte Carlo** radiative transfer.

Monte Carlo: sampling the physics

Instead of solving for I, follow many individual photons and let random numbers decide what happens to each. The basic rule: a variable x with probability density $p(x)$ on $[a,b]$ is sampled from a uniform random number $\zeta \in [0,1)$ by solving

$$\zeta = \int_a^{x_0} p(x)\,\mathrm{d}x.$$

Example: how far does a photon travel? The probability of an interaction between τ and $\tau + \mathrm{d}\tau$ is $p(\tau)\,\mathrm{d}\tau = \mathrm{e}^{-\tau}\mathrm{d}\tau$, so

$$\zeta = 1 - \mathrm{e}^{-\tau} \quad \Longrightarrow \quad \boxed{\tau = -\ln(1-\zeta)}$$

and the physical distance follows from the extinction coefficient along the path. In the same way one samples

- absorption versus scattering (absorbed if $\zeta > \omega_0$);
- the new direction from the phase function (isotropic: $\mu = 2\zeta - 1$, $\phi = 2\pi\zeta'$; Henyey–Greenstein has a closed inverse too);
- the entry direction (diffuse illumination: $\mu = \sqrt{\zeta}$).

The Monte Carlo algorithm

1. **Initialize:** launch a photon packet at the top of the atmosphere with a position and direction drawn from the stellar illumination.
2. **Track:** draw the optical path to the next interaction and move the photon. Decide absorption or scattering. An absorbed photon is "killed": its energy goes into heating and later thermal emission. A scattered photon gets a new direction (and polarization) from the phase function.
3. **Escape:** repeat until the photon leaves the atmosphere; bin it by exit direction and position. The binned counts give the flux (and polarization) versus phase angle and inclination.
4. **Errors:** counts in each bin obey Poisson statistics, so the relative error is $1/\sqrt{N}$; 10^7 or more photons are typical.

- **Strengths:** arbitrary geometry and inhomogeneity (patchy clouds, the full sphere, the terminator in transmission); intuitive.
- **Weakness:** slow in optically thick media, where each photon needs $\sim \tau^2$ steps.

Our slab test used exactly this algorithm with 4×10^4 photons per point.

Putting all the pieces together

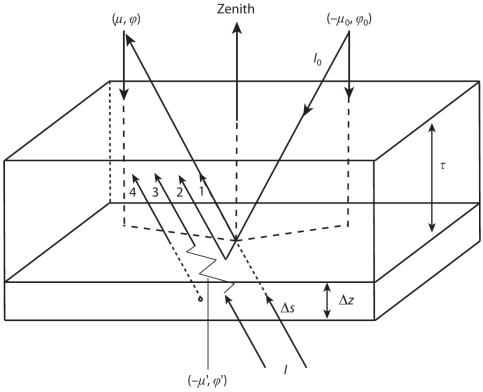


Figure 5: The four processes that change the intensity in a volume element: (1) attenuated starlight, (2) singly scattered starlight, (3) diffusely scattered radiation, (4) thermal emission. Seager (2010), Fig. 6.12, after Liou.

Seager's exercise 7: the full equation for the diffuse intensity,

$$\mu \frac{\mathrm{d}I}{\mathrm{d}\tau} = I - \frac{\omega_0}{4\pi} I_0 \mathrm{e}^{-\tau/\mu_0} P(\mu, \phi; -\mu_0, \phi_0) - \frac{\omega_0}{4\pi} \int_{4\pi} P I \, \mathrm{d}\Omega' - (1 - \omega_0) B.$$

Where each term dominates:

- (1) the direct beam: transit spectroscopy, thin atmospheres;
- (2) single scattering: thin hazes and aerosols ($\tau \ll 1$);
- (3) multiple scattering: thick clouds, Rayleigh-dominated visible light;
- (4) thermal emission: the infrared, where $\omega_0 \approx 0$.

Take-home messages

Simple solutions. Beer's law for pure extinction; isothermal layers exchange flux through $\mathscr{T} = 2E_3(\Delta\tau)$; a linear source function gives Eddington–Barbier. Radiative equilibrium is *local* energy conservation. Clouds add grey extinction and degeneracies; retrieval inverts spectra but must respect physics and chemistry.

Two-stream. Integrate over hemispheres and close with Eddington coefficients. Physics fixes them: $\varepsilon = 1/2$ from the black-body limit, and the hemispheric closure $\gamma_a = \bar{\mu}[2 - \omega_0(1 + g_0)]$, $\gamma_s = \bar{\mu}\omega_0(1 - g_0)$. The Eddington closure makes reflection without scatterers. Accuracy $\sim$ 10–20%.

Deep layers. Diffusion, $F = -(16\sigma_{SB}T^3/3\kappa_R)\,\mathrm{d}T/\mathrm{d}z$, with the harmonic Rosseland mean; Milne's $T^4 = \frac{3}{4}T_{int}^4(\tau_R + C)$. Valid in hot-Jupiter interiors, not in Earth's troposphere.

Next. Temperature–pressure profiles: combining shortwave heating and longwave cooling in the two-stream framework (Heng Ch. 4).

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